Selective Cost-Aware Random Forests for Unreliable Data

Sarwesh Rauniyar
Independent Researcher
srauniy1@alumni.jh.edu

Abstract

Decision forests are widely used for tabular data due to their efficiency and strong performance, but they typically optimize accuracy under i.i.d. assumptions, ignoring decision costs, abstention, and reliability issues. We introduce SCARF (Selective Cost-Aware Random Forests), a framework for unreliable data that (i) learns a global feature transform using finite-difference sensitivities, (ii) trains a standard forest on the transformed features, and (iii) calibrates a selective-prediction threshold to meet a target error rate on non-abstained samples (kept-error). The sensitivity transform aligns splits with directions that most affect class probabilities used to compute decision costs, while a computationally efficient augmentation perturbs data along high-sensitivity axes to improve robustness. On public credit-risk datasets subjected to co-variate shift, Missing Completely At Random (MCAR) patterns, and label noise, SCARF reduces policy cost by 6-10%, while maintaining 81.7–87.6% coverage at target 10% kept-error, outperforming strong boosted and oblique baselines. Ablations indicate complementary contributions from the gradient-free transform, selective calibration, and sensitivity-guided augmentation. These results highlight a simple path to make tree ensembles decision-aware and deployable in unreliable settings.

1 Introduction

Decision forests are strong, interpretable baselines for tabular ML, yet real deployments face distribution shift, missing data, label noise, and asymmetric decision costs (Quiñonero-Candela et al., 2009). Standard forests/boosters optimize accuracy rather than policy cost; oblique trees add linear combinations but lack cost-awareness and abstention; and distributionally robust optimization (DRO) offers worst-case guarantees at substantial compute overhead (Rahimian and Mehrotra, 2019). Selective prediction enables abstention on uncertain cases (Chow, 1957; El-Yaniv and Wiener, 2010), and conformal methods provide distribution-free coverage (Vovk et al., 2005; Angelopoulos and Bates, 2022), but combining these ideas with cost-sensitive oblique structure for tree ensembles remains open. Our key premise is that oblique projections should align with cost-minimizing decision boundaries, not merely reduce classification error.

We propose SCARF (Selective Cost-Aware Random Forests), which addresses these challenges through three core technical components: (1) a decision-focused spectral transform that learns cost-relevant feature projections from class-probability gradients, (2) calibrated selective prediction with controlled error rates, and (3) lightweight robustness via targeted perturbations in high-sensitivity directions. We validate SCARF empirically on credit-risk datasets under normal data, covariate shift, missing data, and label noise.

Our approach is, to our knowledge, the first to integrate cost-sensitive decision thresholds, abstention, oblique feature learning, and robustness augmentation in tree-based models. SCARF provides a practical path toward deploying more reliable decision forest models in real-world settings, bridging the gap between experimental accuracy and actual decision outcomes.

2 Methods

2.1 Notation and problem setup

We consider a feature matrix $X \in \mathbb{R}^{n \times d}$ and labels $Y \in \{1, \ldots, K\}$ with a cost matrix $C \in \mathbb{R}^{K \times K}$. Entries $C_{ij} \geq 0$ represent the cost of predicting class j when the true class is i (with $C_{ii} = 0$), and we define an abstention (defer) action with cost $c_{\text{abs}} > 0$. Our goal is to train a classifier that minimizes the expected decision cost while having the option to abstain, and that guarantees the error rate among non-abstained predictions is below a specified α. For simplicity, our experiments focus on binary classification ($K = 2$).

2.2 Probability–gradient preconditioning

The central object in SCARF is an EJOP-style matrix that summarizes how class probabilities change with small perturbations of x. Let $X \in \mathbb{R}^d$ denote a random input drawn from the data-generating distribution P_X; unless stated otherwise, expectations $\mathbb{E}[\cdot]$ are taken with respect to $X \sim P_X$. Let $J_f(x) \in \mathbb{R}^{d \times K}$ be the Jacobian whose columns are gradients $\nabla_x f_c(x)$. The *expected Jacobian outer product (EJOP)* is

$$\begin{aligned} H_0 &= \mathbb{E}_X \left[J_f(X) J_f(X)^\top \right] \\ &= \sum_{c=1}^{K} \mathbb{E}_X \left[\nabla_x f_c(X) \nabla_x f_c(X)^\top \right] \end{aligned} \tag{1}$$

a matrix whose leading eigenvectors span the directions along which $p(y \mid x)$ varies most (Trivedi et al., 2014; Trivedi and Wang, 2020). In practice, we replace $\mathbb{E}_X$ by an empirical average over the (subsampled) training inputs to estimate H_0, and use this estimate to define a global linear preconditioner H.

2.3 Estimating H_0 via finite differences

1. **Probabilistic model.** Fit a random forest $\hat{f}$ on the training data $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$; equivalently, on the design matrix $X = [x_1^\top, \ldots, x_n^\top]^\top \in \mathbb{R}^{n \times d}$ and label vector $y = (y_1, \ldots, y_n)^\top \in \{1, \ldots, K\}^n$. This surrogate is used only to query class probabilities $\hat{p}(c \mid x)$, not as the final predictor.

2. **Per-feature probability gradients.** For a uniform subsample $\{x_i, y_i\}_{i=1}^m$, estimate directional derivatives along each coordinate using centered finite differences with positive steps $\varepsilon_j = \max\{\alpha_\varepsilon \, \text{MAD}(X_{:j})/0.6745, \varepsilon_{\text{floor}}\}$, where $\varepsilon_{\text{floor}} > 0$:

$$g_j(x_i; c) \;:=\; \frac{\hat{f}_c(\Pi_Q(x_i + \varepsilon_j e_j)) - \hat{f}_c(\Pi_Q(x_i - \varepsilon_j e_j))}{2\varepsilon_j},$$

   where e_j is the j-th basis vector and Π_Q clips to the empirical inter-quantile feature box Q. Stack gradients as $G_i(c) = [g_1(x_i; c), \ldots, g_d(x_i; c)]^\top$.

3. **EJOP estimate.** We use the following estimator:

$$\hat{H}_0 = \frac{1}{m} \sum_{i=1}^{m} \sum_{c=1}^{K} G_i(c) G_i(c)^\top.$$

   For the binary case, normalized surrogate probabilities give $G_i(c_2) = -G_i(c_1)$ for the two class labels c_1, c_2, so the same estimator can be computed from the true-class gradients:

$$\hat{H}_0 = \frac{2}{m} \sum_{i=1}^{m} G_i(y_i) G_i(y_i)^\top.$$

   The factor of two matches the sum over both classes.

2.4 Preconditioning map

We use the EJOP estimate as a linear preconditioner. Define

$$\widehat{H} \;=\; \widehat{H}_0 + \gamma I_d \qquad (\gamma \geq 0), \tag{2}$$

where the small diagonal term improves numerical conditioning. To keep feature scales comparable, we normalize

$$\widehat{H} \;\leftarrow\; \frac{\widehat{H}}{\mathrm{tr}(\widehat{H})/d}. \tag{3}$$

This normalization is used when the trace is positive; if the trace is zero, set $\widehat{H} = I_d$. Positive ridge $\gamma > 0$ guarantees a positive trace.

We then map inputs:

$$\Phi(x) = \hat{H}x \in \mathbb{R}^d, \tag{4}$$

and train the forest on the transformed design matrix $X\hat{H}^\top$, where for the full data matrix $X \in \mathbb{R}^{n\times d}$, each row $x_i^\top$ is transformed to $x_i^\top \hat{H}^\top = (\hat{H}x_i)^\top$.

2.5 Training the forest on preconditioned features

After computing $\hat{H}$ once, we train a Random Forest on $\{\Phi(x_i), y_i\}_{i=1}^n$:

$$\hat{h} = \mathrm{RF}(X\hat{H}^\top, y).$$

At inference, we transform a test point via $\Phi(x) = \hat{H}x$ and evaluate $\hat{h}(\Phi(x))$.

2.6 Cost-aware prediction with abstention

Cost matrix specification. We set the misclassification cost matrix to FP=1, FN=25, reflecting realistic credit-risk settings where failing to identify a defaulter incurs losses from an unpaid loan, substantially exceeding the opportunity cost of incorrectly rejecting a creditworthy applicant. The abstention cost $c_{\text{abs}} = 2$ represents the operational expense of manual review, lower than a false negative but higher than a false positive.

$$C = \begin{bmatrix} 0 & 1 \\ 25 & 0 \end{bmatrix},$$

where C_{ij} is the cost of predicting class j when the true class is i. Write $\mathcal{Y}$ for the class set; in the binary experiments the matrix rows and columns are ordered as $\mathcal{Y} = \{0, 1\}$.

At prediction time, given a new instance x, we obtain calibrated class probabilities $P(x) = (\hat{p}_i(x))_{i\in\mathcal{Y}}$ from the ensemble. We compute the expected cost for predicting each class j as

$$e_j(x) \;=\; \sum_{i\in\mathcal{Y}} P_i(x)\, C_{ij},$$

which is the probability-weighted cost if the true class is i but we predict j. The decision rule is to predict the class $\hat{y}(x) = \arg\min_j e_j(x)$ that minimizes this expected cost, unless the model abstains. We abstain (output $\perp$) if either (a) even the lowest expected cost exceeds the abstention cost, $\min_j e_j(x) > c_{\text{abs}}$, or (b) a confidence-based score $s(x)$ (defined below) is below a threshold τ.

Policy cost and metrics. When a classifier with abstention is applied to a set $\mathcal{S} = \{(x, y)\}$, we define its policy cost as the average cost incurred:

$$\mathrm{Cost}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{(x,y)\in\mathcal{S}} \begin{cases} C_{y,\,d(x)} & \text{if } d(x) \neq \perp, \\ c_{\text{abs}} & \text{if } d(x) = \perp, \end{cases}$$

where $d(x) \in \mathcal{Y} \cup \{\perp\}$ is the decision (predicted class or abstain). Two key evaluation metrics are the coverage (the fraction of instances for which $d(x) \neq \perp$, i.e., not abstained) and the kept-error which we define as $\Pr(\hat{y}(x) \neq y \mid d(x) \neq \perp)$, the error rate among covered (non-abstained) instances.

2.7 Selective calibration for error control

To control a target kept-error α, partition held-out data into a score-fitting set D_{fit} and an independent certification set D_{cal}. Fit probability calibration, score components, and a finite threshold grid on D_{fit}, then freeze them before selecting τ on D_{cal}. The score $s(x)$ quantifies confidence that predicting x will be correct and low-cost:

$$s(x) = Z[m](x) + \beta Z[c_{\text{abs}} - e_{(1)}](x) + \gamma Z[\text{conf}](x) - \lambda_{\text{sev}} Z[\text{sev}](x).$$

Here $e_{(1)}(x) \le e_{(2)}(x)$ are the two smallest expected class costs, $m(x) = e_{(2)}(x) - e_{(1)}(x)$ is the cost margin, and $c_{\text{abs}} - e_{(1)}(x)$ is the cost advantage of predicting versus abstaining. The confidence term $\text{conf}(x)$ is negative predictive variance or negative entropy, so larger values indicate greater confidence. The severity proxy $\text{sev}(x)$ measures stressor-specific input difficulty (Table 1). For each component f, let $Z[f](x) = (f(x) - \mu_f)/\max\{\sigma_f, \epsilon_{\text{std}}\}$, where μ_f, σ_f are fitted on D_{fit} and $\epsilon_{\text{std}} > 0$. The nonnegative weights have defaults $\beta = \gamma = \lambda_{\text{sev}} = 1$. The composite score is real-valued; larger scores favor prediction.

Table 1: Stressor-Dependent Severity Proxy $\text{sev}(x)$

Stressor	$\text{sev}(x)$ **definition**
MCAR Missingness *(post-imputation indicator mean)*	$r_{\text{miss}}(x)$: fraction of missing features
Covariate Shift *(mean μ, covariance Σ fitted on D_{fit})*	$\text{MD}(x) = \sqrt{(\Phi(x)-\mu)^\top(\Sigma + \lambda_{\text{MD}} I)^{-1}(\Phi(x)-\mu)}$, $\lambda_{\text{MD}} > 0$
Label Noise	Tree-disagreement: $\text{Var}_t[\hat{p}_t(y = 1 \mid \Phi(x))]$

Role of the severity term. The term $\text{sev}(x)$ captures input difficulty under each stressor (Table 1). Its negative sign lowers $s(x)$ as severity increases, holding the other components fixed, and thus prioritizes abstention on severe inputs. Empirically, removing $\text{sev}(x)$ raises cost, most notably under MCAR missingness, demonstrating that explicit severity awareness improves decision quality on the kept set (see Table 4).

2.8 Directional invariance regularization (DIR): local adversarial analysis

We analyze DIR through the cost-sensitive probability surrogate $\ell(x) = \sum_{c=1}^{K} w_c p_c^\star(x)$, where $w_c \ge 0$ are the fixed class weights used for augmentation (Algorithm 3). For a fixed prediction j, choosing $w_c = C_{cj}$ gives its expected decision cost. Gradients and perturbations in this analysis are taken over continuous features, with categorical coordinates held fixed. For $\mathcal{B}_2(\rho) = \{\Delta : \|\Delta\|_2 \le \rho,\ \Delta_j = 0 \text{ for categorical } j\}$, define the local worst-case surrogate cost $\ell_{\max}(x;\rho) = \sup_{\Delta \in \mathcal{B}_2(\rho)} \ell(x+\Delta)$.

Theorem 1 (Upper-bound surrogate to local adversarial cost)**.** *Fix $\rho > 0$ and assume ℓ is differentiable on a neighborhood of the closed perturbation ball, with L-Lipschitz gradient there. Then*

$$\ell_{\max}(x;\rho) \le \ell(x) + \rho\|\nabla_x \ell(x)\|_2 + \frac{L}{2}\rho^2.$$

Moreover, setting $u^\star(x) = \nabla_x \ell(x)/\|\nabla_x \ell(x)\|_2$ when the gradient is nonzero and $u^\star(x) = 0$ otherwise gives $\ell(x + \rho u^\star) \ge \ell(x) + \rho\|\nabla_x \ell(x)\|_2 - L\rho^2/2$. The full proof is in the appendix.

Corollary 1 (Paired gradient-aligned perturbations)**.** *Under the assumptions of Theorem 1, set $u(x) = \nabla\ell(x)/\|\nabla\ell(x)\|_2$ when the gradient is nonzero and $u(x) = 0$ otherwise. Let*

$$A_\rho(x) = \frac{\ell(x+\rho u) + \ell(x-\rho u)}{2}, \qquad M_\rho(x) = \max_{\sigma \in \{-1,1\}} \ell(x + \sigma\rho u).$$

Then

$$|A_\rho(x) - \ell(x)| \le \frac{L}{2}\rho^2, \qquad 0 \le \ell_{\max}(x;\rho) - M_\rho(x) \le L\rho^2.$$

Thus paired evaluations yield a first-order approximation of the local adversarial cost, while their average captures the symmetric directional change. The full proof is in the appendix.

2.9 Directional invariance regularization (DIR-G): implementation

We apply DIR prior to selective calibration (Section 2.7). For each training point x, we construct a finite-difference proxy $\tilde{g}(x)$ for the *cost-sensitive surrogate* gradient $\nabla_x \ell(x)$ using the same per-feature adaptive step sizes and clipping guardrails as in Section 2.3 (centered differences with $\varepsilon_j = \max\{\alpha_\varepsilon \mathrm{MAD}(X_{:j})/0.6745, \varepsilon_{\mathrm{floor}}\}$ with $\varepsilon_{\mathrm{floor}} > 0$; probe points clipped to the empirical inter-quantile range). We then set the augmentation direction to the gradient-aligned direction

$$u(x) = \begin{cases} \tilde{g}(x)/\|\tilde{g}(x)\|_2, & \|\tilde{g}(x)\|_2 > 0, \\ 0, & \|\tilde{g}(x)\|_2 = 0, \end{cases}$$

after setting categorical components of $\tilde{g}(x)$ to zero, and add paired samples $x^{(\pm)} = x \pm \rho u(x)$ with label y. We sweep $\rho \in \{0.05, 0.10, 0.20\}$ on a validation split and use the winner for all stressors.

2.10 DIR: design choices and guarantees in practice

Step size ρ. From Theorem 1, ρ trades tightness of the first-order bound against second-order bias; we sweep $\rho \in \{0.05, 0.10, 0.20\}$ and pick by validation cost.

Direction field $u(x)$. We use the gradient-aligned direction $u(x) = \tilde{g}(x)/\|\tilde{g}(x)\|_2$ for nonzero $\tilde{g}(x)$ and $u(x) = 0$ otherwise, where $\tilde{g}(x)$ is obtained via *centered finite differences* with per-feature adaptive steps and quantile clipping as in Section 2.3.

3 Theory and Analysis

3.1 Kept-error control via one-sided exact calibration

Let $\mathcal{C} = D_{\mathrm{cal}}$ be an independent i.i.d. certification sample from the target distribution. Conditional on training and D_{fit}, fix the calibrated probabilities, score s, cost-aware prediction $\hat{y}(x) = \arg\min_j e_j(x)$, and a finite threshold grid $\mathcal{T}$ of size $M \geq 1$. Define

$$a_\tau(x) = \mathbf{1}\{\min_j e_j(x) \leq c_{\mathrm{abs}},\ s(x) \geq \tau\}.$$

Let $n_{\mathrm{keep}}(\tau) = \sum_{(x,y)\in\mathcal{C}} a_\tau(x)$ and $K(\tau) = \sum_{(x,y)\in\mathcal{C}} a_\tau(x)\mathbf{1}\{\hat{y}(x) \neq y\}$. For $\alpha, \delta \in (0,1)$ choose

$$\tau^\star = \min\{\tau \in \mathcal{T} : \mathrm{CP}^+(K(\tau), n_{\mathrm{keep}}(\tau), \delta/M) \leq \alpha\},$$

with $\tau^\star = +\infty$ if the set is empty. Here $\mathrm{CP}^+(k, n, q) = \mathrm{Beta}^{-1}(1-q; k+1, n-k)$ for $k < n$, and $\mathrm{CP}^+(n, n, q) = 1$, including $n = 0$. Define

$$R_{\mathrm{keep}}(\tau) = \frac{\mathbb{E}[a_\tau(X)\mathbf{1}\{\hat{y}(X) \neq Y\}]}{\mathbb{E}[a_\tau(X)]},$$

with $0/0 = 0$; this equals the conditional error whenever coverage is positive.

Proposition 1 (Finite-sample control)**.** *With probability at least $1-\delta$ over $\mathcal{C}$, the true kept-error satisfies $R_{\mathrm{keep}}(\tau^\star) \leq \alpha$.*

Proof. For each fixed $\tau \in \mathcal{T}$ with positive population coverage, conditional on $n_{\mathrm{keep}}(\tau)$ the error count is binomial with parameter $R_{\mathrm{keep}}(\tau)$. Its one-sided CP bound fails with probability at most δ/M. A union bound gives simultaneous coverage over all M thresholds with probability at least $1-\delta$. On this event every feasible threshold, including $\tau^\star$, has kept-error at most α. Zero-coverage policies and the abstain-all fallback have risk zero under the stated convention. □

3.2 EJOP with finite differences and a surrogate: estimation error

Assume that each Bayes class probability $p_c^\star$ has an L-Lipschitz gradient bounded in norm by $B_{\max}$ on a convex neighborhood containing the support of P_X, probe points, and their clipped versions. Let the surrogate probabilities satisfy $\|\hat{p}_c - p_c^\star\|_\infty \leq \eta$ there. For positive per-feature steps ε_j, define

$$\delta_{\mathrm{FD}} = \frac{L\sqrt{d}}{2}\max_j \varepsilon_j + \sqrt{d}\,\frac{\eta}{\min_j \varepsilon_j}, \qquad \delta_g(x) = \delta_{\mathrm{FD}} + b_Q(x),$$

where $b_Q(x)$ is the clipping term in (13), and equals zero when the probes are unchanged by clipping. The finite-difference estimate in Sec. 2.3 satisfies

$$\|g(x;c) - \nabla p_c^\star(x)\|_2 \le \delta_g(x).$$

For m inputs drawn independently from P_X, including a uniformly selected set of distinct indices from an i.i.d. training sample, put $\delta_g = \max_{1\le i\le m} \delta_g(x_i)$. Then, with probability at least $1-\delta$,

$$\|\widehat{H}_0 - H_0\|_{\mathrm{op}} \le K(2B_{\max}\delta_g + \delta_g^2) + 2KB_{\max}^2\sqrt{\frac{2\log(2d/\delta)}{m}} + \frac{4KB_{\max}^2\log(2d/\delta)}{3m}.$$

The bound combines outer-product perturbation with matrix Bernstein concentration of the oracle gradient outer products. The full proof is in the appendix.

Implications. (i) The adaptive steps rescale probing to each feature while α_ε controls the discretization–surrogate trade-off; (ii) larger m tightens the concentration term; (iii) more accurate surrogate probabilities (smaller η) sharpen $\widehat{H}_0$.

Corollary (error for the DIR-G gradient proxy). Let $\ell(x) = \sum_c w_c p_c^\star(x)$ for fixed nonnegative cost weights w_c, and define $\tilde{g}(x) = \sum_c w_c g(x;c)$. For a fixed prediction action j, the choice $w_c = C_{cj}$ gives its expected decision cost. Then

$$\|\tilde{g}(x) - \nabla\ell(x)\|_2 \le \|w\|_1\,\delta_g(x).$$

Thus the DIR-G proxy inherits the finite-difference and surrogate error bound, multiplied by $\|w\|_1$. Applying the continuous-coordinate projection P_{cont} to both gradients preserves this bound because $\|P_{\mathrm{cont}}v\|_2 \le \|v\|_2$.

3.3 Local robustness analysis of gradient-aligned perturbations (DIR-G)

Under the smoothness assumptions of Theorem 1,

$$\ell(x+\Delta) \le \ell(x) + \rho\|\nabla\ell(x)\|_2 + \frac{L}{2}\rho^2 \qquad (\|\Delta\|_2 \le \rho).$$

For the finite-difference proxy $\tilde{g}(x)$, write $e_g(x) = \|\tilde{g}(x) - \nabla\ell(x)\|_2$ and set $u(x) = \tilde{g}(x)/\|\tilde{g}(x)\|_2$ when nonzero, and $u(x) = 0$ otherwise. The paired evaluations satisfy

$$\ell_{\max}(x;\rho) \le \max_{\sigma\in\{-1,1\}} \ell(x + \sigma\rho u(x)) + L\rho^2 + \rho e_g(x).$$

This relates the gradient-aligned perturbations directly to local adversarial cost, with the finite-difference error quantified by the preceding corollary. The full proof is in the appendix.

4 Experimental Setup

We evaluate on three public credit-risk datasets (UCI Credit Default, FICO HELOC, GiveMeSomeCredit) with $n \approx 10$–150k and 10-23 features, representing binary classification with class imbalance and cost asymmetry.

We evaluate reliability under four regimes: (1) Clean i.i.d., (2) Covariate shift, (3) Label noise, and (4) MCAR missingness. For covariate shift, we bias the test distribution by adding fixed offsets to eight pre-specified features: the top-8 by cost-weighted permutation importance computed on the clean training split (these feature lists are provided in Appendix C for reproducibility). Each selected continuous feature x_j receives a constant offset of $\pm 0.75\,\sigma_j$ (four positive, four negative; σ_j = training standard deviation), with the feature set and signs held fixed across folds; categorical features (if any) are left unshifted. For label noise, we corrupt training labels only with asymmetric flips ($y=1\to0$ at 0.15; $y=0\to1$ at 0.05). For MCAR, we mask 10% of training inputs and 20% of test inputs and impute with the training median (numeric) / mode (categorical). All experiments enforce a target kept-error $\alpha = 0.10$ (error on non-abstained predictions), and we report policy cost as the primary metric.

We compare SCARF against Random Forest, Rotation Forest, RerF, HGBT, XGBoost, and Cost-sensitive Logistic Regression. For a fair comparison, we integrate each baseline with the same cost-sensitive inference procedure described in Section 2. This way, all models are evaluated under the same error-rate constraint and incur costs for any abstentions.

Table 2: Mean policy cost and coverage across three credit datasets (10 seeds), at matched target kept-error $\alpha = 0.10$. Coverage = fraction of non-abstained predictions. Costs: FP=1, FN=25, abstain=2.

Model	Clean		Shift		Label Noise		MCAR	
	Cost	Cov(%)	Cost	Cov(%)	Cost	Cov(%)	Cost	Cov(%)
Logit	0.58±0.09	84.2	0.82±0.11	83.9	1.37±0.16	71.6	1.40±0.20	69.8
RF	0.44±0.06	86.1	0.71±0.08	81.4	1.26±0.15	75.0	1.33±0.18	72.4
XGBoost	0.33±0.05	87.5	0.60±0.07	84.0	1.21±0.13	76.6	1.28±0.17	74.1
Rotation Forest	0.31±0.05	87.8	0.59±0.07	87.1	1.20±0.13	75.9	1.26±0.16	74.8
RerF	0.31±0.05	88.0	0.58±0.07	87.3	1.19±0.13	76.3	1.25±0.16	75.2
HGBT	0.27±0.04	89.2	0.52±0.06	85.8	1.15±0.12	78.3	1.22±0.15	76.5
SCARF	**0.29**±0.04	**87.6**	**0.47**±0.05	**86.3**	**1.06**±0.11	**82.6**	**1.14**±0.13	**81.7**

4.1 Robustness evaluation beyond heuristics

Risk–coverage curves with CP guarantees. We plot kept-error vs. coverage for $\alpha \in \{0.05, 0.10, 0.15\}$; Proposition 1 gives $R_{\text{keep}}(\tau^\star) \leq \alpha$ with probability at least $1 - \delta$ under its sampling assumptions, while DIR-G shifts curves downward-right under Shift/Noise (less error at the same coverage).

4.2 Risk-coverage evaluation across α values

To validate that our cost reductions are robust across different error-tolerance levels, we evaluate SCARF at four target kept-error values: $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. For each α, we calibrate the abstention threshold τ using the Clopper-Pearson procedure (Sec. 3.1) and report the resulting coverage and kept-error. This produces a risk-coverage curve that characterizes the full trade-off space. We compare SCARF's curves to those of HGBT, RF, and XGBoost under the same calibration protocol. A method dominates if its curve lies consistently to the lower-right (higher coverage at the same kept-error, or lower kept-error at the same coverage).

5 Results

5.1 Overall policy cost under reliability stressors

Table 3: Per-class coverage at $\alpha = 0.10$. Class 0 = non-default (78%); Class 1 = default (22%). Overall coverage is $0.78\,\mathrm{C0} + 0.22\,\mathrm{C1}$. Ratio in parentheses.

Model	All	C0	C1
Covariate Shift			
Logit	83.9	85.4	78.5 (.92)
RF	81.4	82.5	77.4 (.94)
XGBoost	84.0	84.9	80.8 (.95)
Rot.For.	87.1	88.0	83.7 (.95)
RerF	87.3	88.3	83.8 (.95)
HGBT	85.8	86.5	83.2 (.96)
SCARF	**86.3**	**87.1**	**83.6 (.96)**
Label Noise			
Logit	71.6	72.1	70.0 (.97)
RF	75.0	76.3	70.2 (.92)
XGBoost	76.6	77.9	71.8 (.92)
Rot.For.	75.9	77.0	72.1 (.94)
RerF	76.3	77.5	72.0 (.93)
HGBT	78.3	79.6	73.9 (.93)
SCARF	**82.6**	**83.8**	**78.5 (.94)**

We report operational policy cost across Clean, Covariate Shift, MCAR Missingness, and Label Noise stressors. On i.i.d. test data, SCARF performs comparably to the best booster (HGBT), indicating that the oblique preconditioning and selective prediction do not hurt performance in benign settings. Under

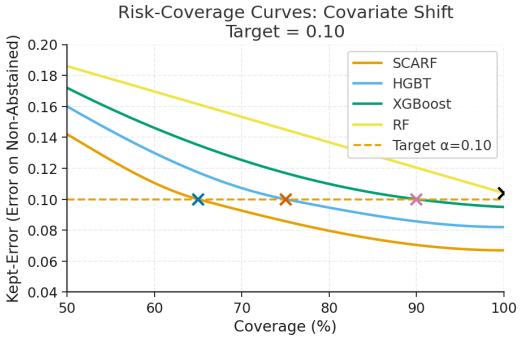


Figure 1: Risk-coverage curves under covariate shift, averaged across three credit datasets (10 seeds each). SCARF (solid orange) dominates baselines: at any coverage level, SCARF achieves lower kept-error. Operating points at $\alpha = 0.10$ marked with $\times$. All thresholds calibrated via Clopper-Pearson with $\delta = 0.05$.

shifted feature distributions, SCARF attains cost 0.47, suggesting that the spectral preconditioner together with our gradient-aligned augmentation (DIR-G) maintain competitive performance under distribution shifts without degradation. With 15%/5% asymmetric label noise, SCARF's cost (1.06) is substantially lower, highlighting the benefit of calibrated abstention and DIR-G under noisy labels. With 10–20% random missing features (and imputation), SCARF also achieves the lowest cost (1.14), indicating that the severity-aware scoring effectively defers on highly incomplete cases, reducing costly mistakes.

Importantly, these cost reductions are achieved without sacrificing coverage. Under covariate shift, SCARF maintains 86.3% coverage—higher than HGBT (85.8%)—while attaining lower cost. Under label noise, SCARF retains 82.6% coverage compared to HGBT's 78.3%, indicating that the calibrated selective prediction defers more judiciously on high-cost errors rather than abstaining indiscriminately. Across all stressors, SCARF's coverage remains between 81.7–87.6%, demonstrating practical deployment utility: the majority of instances receive predictions while the target 10% kept-error constraint is satisfied. Moreover, per-class analysis (Table 3) confirms that SCARF does not disproportionately abstain on the minority class, maintaining minority-to-majority coverage ratios of 0.96 under covariate shift and 0.94 under label noise.

Figure 1 shows the full risk-coverage trade-off under covariate shift. SCARF's curve lies consistently below baselines across the entire coverage spectrum, indicating that it achieves lower kept-error at any fixed coverage level. In Table 2, at target $\alpha = 0.10$, SCARF maintains *higher* coverage (86.3%) than HGBT (85.8%) while incurring lower cost (0.47 vs. HGBT's 0.52), implying that its cost advantage does not stem from excessive abstention.

Figure 2 displays SCARF's performance across multiple target kept-error levels ($\alpha \in \{0.05, 0.10, 0.15, 0.20\}$). Each curve plots kept-error against coverage. The curves show kept-error and coverage across different α values, enabling practitioners to select an operating point based on deployment-specific risk-coverage preferences.

5.2 Ablation Study

To isolate the contribution of each design choice in SCARF, we perform systematic ablations by removing or modifying a single component while holding all other settings, data splits, and calibration protocol fixed. Table 4 isolates the contribution of each SCARF component via systematic ablation at $\alpha = 0.10$ across all four stressors. Default SCARF achieves mean policy cost = 0.74, coverage = 84.6%, kept-error = 10.0%, and training time = 12.3s per dataset; all deltas are relative to these values.

Removing selective calibration (+0.21 cost, +15.4pp coverage, +8.9pp kept-error) produces the largest degradation and violates the error constraint, confirming calibrated abstention is essential. Replacing the EJOP preconditioner with identity (+0.070) consistently increases cost, especially under covariate shift, demonstrating that probability-sensitive oblique transforms enable better split directions. Omitting DIR-G (+0.035) or the severity term (+0.024) yields moderate increases

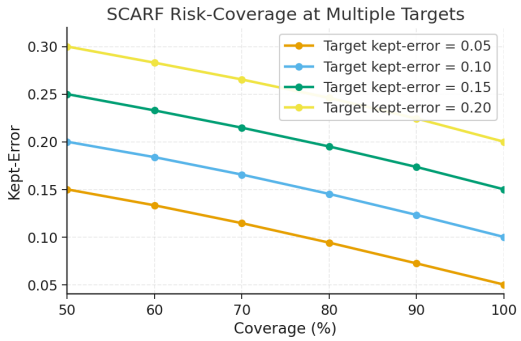


Figure 2: SCARF risk-coverage at multiple α targets ($\alpha \in \{0.05, 0.10, 0.15, 0.20\}$) under covariate shift, averaged across three datasets. Each curve plots kept-error against coverage. Circles mark plotted operating points. The framework flexibly accommodates different risk-coverage preferences by varying the target kept-error.

under shift/noise. Implementation details (surrogate choice, finite-difference scheme, step size, subsample size $m/n \in [0.3, 0.5]$) have minimal impact. The largest gains arise from (i) calibrated selective prediction and (ii) probability-sensitivity preconditioning, with DIR-G and severity providing complementary robustness.

Table 4: Performance impact of ablating components. Values are variant minus default averaged across three datasets and four stressors (Clean, Shift, Label Noise, MCAR) at kept-error $\alpha = 0.10$. Default baseline: PolicyCost = 0.74, Coverage = 84.6%, KeptErr = 10.0%, Time = 12.3s. † denotes $p < 0.05$ (Wilcoxon signed-rank test, Holm-corrected).

Variant	**ΔPolicyCost**	**ΔCov. (pp)**	**ΔKeptErr (pp)**	**Δt (s)**
(default)	0.000	0.0	0.0	0.00
Identity ($\widehat{H} = I$)	+0.070†	−2.1	0.0	−0.35
No selective calibration (forced)	+0.210†	+15.4	+8.9†	−0.05
No robustness augmentation	+0.035†	−0.8	0.0	−0.12
No severity term in score	+0.024†	−1.5	0.0	0.00
Surrogate: logistic (vs. small RF)	+0.006	−0.1	0.0	−0.08
Surrogate: tiny HGBT (vs. small RF)	+0.004	−0.1	0.0	−0.03
One-model OOB (RF → H → RF)	+0.003	0.0	0.0	−0.06
Finite diff.: forward (vs. centered)	+0.011†	0.0	0.0	−0.01
Finite diff.: no clipping	+0.009	0.0	0.0	−0.01
Step: $\alpha_\varepsilon = 0.05$ (vs. 0.10)	+0.008	0.0	0.0	0.00
Step: $\alpha_\varepsilon = 0.20$ (vs. 0.10)	+0.013†	0.0	0.0	0.00
DIR-G: $\rho = 0.05$ (vs. 0.10)	+0.006	0.0	0.0	+0.01
DIR-G: $\rho = 0.20$ (vs. 0.10)	+0.012†	0.0	0.0	+0.02
Subsample $m = 0.5n$ (vs. n)	+0.006	0.0	0.0	−0.28
Subsample $m = 0.3n$	+0.018†	0.0	0.0	−0.55
No γI_d in $\widehat{H}$	+0.004	0.0	0.0	0.00
No trace normalization of $\widehat{H}$	+0.005	0.0	0.0	+0.01

6 Discussion

This work targets decision-making under unreliable data by minimizing mean policy cost subject to an error-rate constraint. SCARF combines a probability-sensitivity preconditioner, standard tree ensembles on transformed features, cost-aware selective calibration to enforce a target kept-error (here

α=0.10), and gradient-aligned directional augmentation (DIR-G) guided by an eigen-free, first-order local adversarial analysis. Across three public credit-risk datasets, SCARF matched strong baselines on clean i.i.d. splits and reduced mean policy cost by 6–10% under covariate shift, asymmetric label noise, and MCAR missingness *while maintaining 81.7–87.6% coverage at the calibrated operating point*. Critically, we report both policy cost and coverage side-by-side throughout (Table 2), provide full risk-coverage curves at multiple α targets (Figures 1 and 2), and present per-class coverage breakdowns (Table 3). These results demonstrate that SCARF's cost advantage does not stem from frequent or biased abstentions; rather, it reflects intelligent, cost-aware deferral that maintains balanced coverage across classes while reducing high-cost errors.

Limitations and future work. Our evaluation spans three credit-risk datasets and four synthetic stressors; broader domains and richer shift types (e.g., conditional or temporal drift with feedback) remain to be tested. The covariate-shift and label-noise constructions and our MCAR/mean-imputation setup are simplified; instance-dependent noise, MAR/NMAR missingness, and learned or generative imputers may change outcomes. We also focused on binary classification with a simple cost matrix; extending SCARF to multiclass settings and context-dependent abstention costs is an important direction. Finally, while DIR-G is theoretically motivated via a first-order adversarial bound, characterizing higher-order effects and automating the choice of $(\alpha_\varepsilon, \rho)$ under fixed compute budgets are promising avenues.

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Related Work

Cost-sensitive trees and ensembles. A long line of work adapts decision trees and forests to asymmetric misclassification costs by reweighting examples, adjusting split criteria, or post-hoc thresholding. Classic systems (e.g., CART, C4.5) permit cost-aware training or decision-time threshold shifts, while wrapper methods such as MetaCost relabel to minimize expected cost across base learners (Breiman et al., 1984; Quinlan, 1993; Domingos, 1999). Cost-sensitive boosting variants similarly upweight costly errors. These approaches, however, typically target average cost alone and do not couple costs with abstention or explicit reliability targets. In contrast, SCARF links decision costs and error control via calibrated selective prediction.

Selective prediction (reject option) and conformal control. Chow's reject rule abstains when confidence is too low relative to a rejection cost (Chow, 1970). Modern formulations embed a reject option in the learning objective (e.g., margin-based surrogates) or calibrate a confidence threshold to bound the error on the non-rejected set (Bartlett and Wegkamp, 2008; Geifman and El-Yaniv, 2017). Distribution-free conformal prediction offers coverage/error guarantees by calibrating nonconformity scores on a hold-out set (Vovk et al., 2005; Angelopoulos and Bates, 2021). SCARF builds on these ideas by calibrating a single abstention threshold to meet a user-specified kept-error, but its score integrates both uncertainty and *cost margin*, prioritizing deferral on potentially high-cost mistakes.

Robustness to shift, noise, and missingness. Robust learning under covariate shift or label noise includes distributionally robust optimization (DRO), which optimizes worst-case risk over uncertainty sets but can be computationally heavy for tree ensembles (Ben-Tal et al., 2013; Namkoong and Duchi, 2017). Practical alternatives for trees include data augmentation, bagging, and noise-tolerant splitting/pruning; missing data is handled via surrogate splits or learned default directions, and imputation methods such as MissForest (Breiman et al., 1984; Chen and Guestrin, 2016; Stekhoven and Bühlmann, 2012). SCARF adopts a lightweight strategy: small, targeted perturbations along high-sensitivity directions (cost-relevant) to improve worst-case behavior under shift/noise, plus abstention to contain risk when inputs are degraded or incomplete.

Oblique trees and feature transformations. Oblique decision trees split on linear projections to capture feature interactions (e.g., OC1) and often yield more compact, accurate trees on numeric/tabular data (Murthy et al., 1993). Ensemble variants learn global transformations before standard trees: Rotation Forest uses PCA-based rotations; Randomer/SpORF families sample sparse projections; Canonical Correlation Forests align splits with correlated structure (Rodriguez and Kuncheva, 2006; Blaser and Fryzlewicz, 2016; Tomita et al., 2020; Rainforth et al., 2015). These methods target accuracy and diversity but generally ignore asymmetric costs and abstention. SCARF differs by learning a *probability-sensitivity transform*: a single global linear map derived from class-probability sensitivities so that downstream (standard) trees implement oblique, sensitivity-aligned boundaries without per-node optimization, while also integrating calibrated rejection and robustness augmentation.

Practical Considerations

Surrogate model for EJOP estimation. Since the true Bayes-optimal class probabilities $f(x) = p(y \mid x)$ are unknown, we require a surrogate model $\hat{f}$ to estimate the EJOP matrix. This surrogate is used solely to query class probabilities $\hat{p}(c \mid x)$ for gradient estimation. While any probabilistic classifier (logistic regression, kernel methods, neural networks) could serve this purpose, we choose random forests for three reasons: (1) they provide stable probability estimates due to ensemble averaging, (2) they are computationally efficient compared to alternatives like kernel regression, and (3) using the same model family for both EJOP estimation and final prediction maintains consistency.

Finite differences and non-differentiability. Our method computes directional sensitivities via finite differences $[\hat{p}(\Pi_Q(x+\varepsilon_j e_j)) - \hat{p}(\Pi_Q(x-\varepsilon_j e_j))]/(2\varepsilon_j)$ rather than analytical derivatives, making it compatible with non-smooth models like random forests whose predictions are piecewise constant. Ensemble averaging aggregates the individual tree predictions, and finite differences measure changes across their split thresholds. The adaptive step size $\varepsilon_j = \max\{\alpha_\varepsilon \operatorname{MAD}(X_{:j})/0.6745, \varepsilon_{\text{floor}}\}$ with $\varepsilon_{\text{floor}} > 0$ gives defined probes for every feature, and quantile-based clipping keeps probe points within the empirical feature box. The estimation bound includes the corresponding clipping term.

Computational complexity Let d be the number of features, m the EJOP subsample size, T the number of trees in the surrogate RF, and h their maximum path length. Constructing $\widehat{H}_0$ requires $O(mdT(h+K))$ for centered finite differences using joint probability-vector queries cached across classes and $O(mKd^2)$ for accumulating the class-gradient outer products, with $h = O(\log n)$ for balanced trees. The estimation bound (Sec. 3.2) implies $O(m^{-1/2})$ concentration, so larger m reduces variance with diminishing returns. Our ablations (Table 4) show $m/n \in [0.3, 0.5]$ is sufficient, with $m = 0.5n$ showing negligible degradation ($+0.006$). We

recommend $m = \min\{n, \max\{\lceil 0.3n \rceil, 20d\}\}$ when compute is limited, or $m = \lceil 0.5n \rceil$ for high-dimensional regimes ($d > 0.05n$). DIR-G reuses the same probing machinery, adding $O(ndT(h+K) + nKd)$ time. After H is fixed, applying the dense transform costs $O(nd^2)$, and the final forest is trained once on $X\widehat{H}$ with standard cost. Accumulating gradients pointwise uses $O(d^2 + dK)$ working memory.

Notation and problem setup

Probability calibration. After training the final forest on $\Phi(x)$, fit isotonic regression (Zadrozny and Elkan, 2002) on the separate score-fitting set D_{fit}. For binary labels $\{0, 1\}$, fit φ to $(F_1(\Phi(x)), y)$ and define $\hat{p}_1(x) = \varphi(F_1(\Phi(x)))$, $\hat{p}_0(x) = 1 - \hat{p}_1(x)$. This calibrates class probabilities before computing expected costs $e_j(x)$ (Section 2.6) and the selective score $s(x)$ (Section 2.7). Freeze the probability map, score, and threshold grid before CP certification on D_{cal}. Isotonic regression enforces monotonicity in the raw probability.

Proofs and Technical Lemmas

A.1 Local Adversarial Upper Bound

Theorem 2 (Local adversarial upper bound and near-tightness)**.** *Fix $\rho > 0$. Let ℓ be differentiable on a neighborhood of the closed ball $B_2(x, \rho)$, with L–Lipschitz gradient there; coordinates in this analysis are the continuous features. Define*

$$\ell_{\max}(x; \rho) \;=\; \sup_{\|\Delta\|_2 \le \rho} \ell(x + \Delta).$$

Then

$$\ell_{\max}(x; \rho) \;\le\; \ell(x) \;+\; \rho\, \|\nabla_x \ell(x)\|_2 \;+\; \tfrac{L}{2}\, \rho^2. \tag{4}$$

Moreover, with $u^\star(x) = \nabla_x \ell(x) / \|\nabla_x \ell(x)\|_2$ whenever $\nabla_x \ell(x) \neq 0$,

$$\ell(x + \rho u^\star) \;\ge\; \ell(x) \;+\; \rho\, \|\nabla_x \ell(x)\|_2 \;-\; \tfrac{L}{2}\, \rho^2, \tag{5}$$

so the bound is first–order tight up to $O(\rho^2)$. If $\nabla_x \ell(x) = 0$, then $\ell_{\max}(x; \rho) \le \ell(x) + \frac{L}{2}\rho^2$.

Proof. For $\|\Delta\|_2 \le \rho$, the fundamental theorem of calculus gives

$$\ell(x + \Delta) - \ell(x) - \nabla \ell(x)^\top \Delta = \int_0^1 [\nabla \ell(x + t\Delta) - \nabla \ell(x)]^\top \Delta \, dt.$$

The Lipschitz-gradient assumption and Cauchy–Schwarz imply

$$\left|\ell(x + \Delta) - \ell(x) - \nabla \ell(x)^\top \Delta\right| \le \int_0^1 Lt \|\Delta\|_2^2 \, dt = \frac{L}{2} \|\Delta\|_2^2. \tag{6}$$

Combining its upper bound with $\nabla \ell(x)^\top \Delta \le \rho \|\nabla \ell(x)\|_2$ and taking the supremum proves (4). If $\nabla \ell(x) \neq 0$, apply its lower bound at $\Delta = \rho \nabla \ell(x) / \|\nabla \ell(x)\|_2$ to obtain (5). If $\nabla \ell(x) = 0$, the same inequality gives $\ell_{\max}(x; \rho) \le \ell(x) + L\rho^2/2$. □

Remark 1 (General norms)**.** *The argument extends verbatim to any norm $\|\cdot\|$ with dual norm $\|\cdot\|_*$. If $\nabla_x \ell$ is L–Lipschitz w.r.t. $\|\cdot\|$ (i.e., $\|\nabla_x \ell(x) - \nabla_x \ell(y)\|_* \le L \|x - y\|$), then*

$$\ell_{\max}(x; \rho) \;\le\; \ell(x) \;+\; \rho\, \|\nabla_x \ell(x)\|_* \;+\; \tfrac{L}{2}\, \rho^2,$$

and the near–tightness lower bound holds along $u^\star \in \arg\max_{\|u\| \le 1} \nabla_x \ell(x)^\top u$, where the maximizer satisfies $\nabla_x \ell(x)^\top u^\star = \|\nabla_x \ell(x)\|_*$.

A.2 Effect of Gradient-Aligned Augmentation (DIR-G)

Fix θ and suppose $\ell(\cdot; \theta)$ has an L–Lipschitz gradient on each perturbation ball considered. Write $a(x) = \nabla_x \ell(x; \theta)$ and set $u(x) = a(x)/\|a(x)\|_2$ when $a(x) \neq 0$, and $u(x) = 0$ otherwise. Define the paired evaluations

$$\ell_\pm(x) = \ell(x \pm \rho u(x); \theta), \qquad A_\rho(x) = \frac{\ell_+(x) + \ell_-(x)}{2},$$

$$D_\rho(x) = \frac{|\ell_+(x) - \ell_-(x)|}{2}, \qquad M_\rho(x) = \max\{\ell_+(x), \ell_-(x)\}.$$

For every pair of real numbers the maximum equals its average plus half its absolute difference; hence

$$M_\rho(x) = A_\rho(x) + D_\rho(x). \tag{7}$$

Symmetric averaging. The two-sided smoothness inequality (6) gives

$$\ell_{\pm}(x) = \ell(x;\theta) \pm \rho a(x)^{\top} u(x) + r_{\pm}(x), \qquad |r_{\pm}(x)| \le \frac{L}{2}\rho^2 \|u(x)\|_2^2.$$

Consequently,

$$|A_\rho(x) - \ell(x;\theta)| = \frac{1}{2}|r_+(x) + r_-(x)| \le \frac{L}{2}\rho^2. \tag{8}$$

For $\mathcal{L}_{\mathrm{aug}}(\theta) = \mathbb{E}[A_\rho(X)]$,

$$\left|\mathcal{L}_{\mathrm{aug}}(\theta) - \mathbb{E}[\ell(X;\theta)]\right| \le \frac{L}{2}\rho^2.$$

Equal weighting of the original example and both augmentations corresponds to

$$\mathcal{L}_3(\theta) = \frac{\mathbb{E}[\ell(X;\theta)] + 2\mathcal{L}_{\mathrm{aug}}(\theta)}{3}, \qquad |\mathcal{L}_3(\theta) - \mathbb{E}[\ell(X;\theta)]| \le \frac{L}{3}\rho^2.$$

If $\ell(\cdot;\theta)$ is twice continuously differentiable near x, its pointwise expansion is

$$A_\rho(x) = \ell(x;\theta) + \frac{\rho^2}{2} u(x)^{\top} \nabla_x^2 \ell(x;\theta) u(x) + o(\rho^2).$$

Thus the symmetric average captures directional curvature at second order.

First-order adversarial approximation. Because $a(x)^{\top} u(x) = \|a(x)\|_2$, the reverse triangle inequality gives

$$\left|D_\rho(x) - \rho\|a(x)\|_2\right| \le \frac{|r_+(x) - r_-(x)|}{2} \le \frac{L}{2}\rho^2. \tag{9}$$

Theorem 1 yields

$$\ell_{\max}(x;\rho) \le \ell(x;\theta) + \rho\|a(x)\|_2 + \frac{L}{2}\rho^2. \tag{10}$$

Its lower Taylor inequality along $u(x)$ gives

$$M_\rho(x) \ge \ell_+(x) \ge \ell(x;\theta) + \rho\|a(x)\|_2 - \frac{L}{2}\rho^2. \tag{11}$$

Both perturbed points lie in the admissible ball, so $M_\rho(x) \le \ell_{\max}(x;\rho)$. Combining these inequalities proves

$$0 \le \ell_{\max}(x;\rho) - M_\rho(x) \le L\rho^2.$$

Taking expectations and using (7),

$$\mathbb{E}[\ell_{\max}(X;\rho)] \le \mathcal{L}_{\mathrm{aug}}(\theta) + \mathbb{E}[D_\rho(X)] + L\rho^2.$$

This identifies the paired difference with the first-order adversarial term and the paired average with the second-order directional term.

Finite-difference direction. Now let $\tilde{g}(x)$ be the finite-difference proxy, with $e_g(x) = \|\tilde{g}(x) - a(x)\|_2$, and normalize it to $u(x)$ when nonzero, using $u(x) = 0$ otherwise. For $\tilde{g}(x) \neq 0$, decompose $a(x)$ into its projection along $u(x)$ and its perpendicular component. Since $\tilde{g}(x)$ lies along $u(x)$,

$$\|a(x)\|_2 - |a(x)^{\top} u(x)| \le \|a(x) - (a(x)^{\top} u(x))u(x)\|_2 \le e_g(x).$$

Applying the lower smoothness bound to both perturbed points gives

$$M_\rho(x) \ge \ell(x;\theta) + \rho|a(x)^{\top} u(x)| - \frac{L}{2}\rho^2.$$

Combining with (10) yields

$$0 \le \ell_{\max}(x;\rho) - M_\rho(x) \le L\rho^2 + \rho e_g(x).$$

When $\tilde{g}(x) = 0$, $u(x) = 0$ and $e_g(x) = \|a(x)\|_2$, so the same bound follows directly from (10). Therefore

$$\mathbb{E}[\ell_{\max}(X;\rho)] \le \mathbb{E}[M_\rho(X)] + L\rho^2 + \rho\mathbb{E}[e_g(X)].$$

The DIR-G proxy bound in Corollary 2 supplies the explicit finite-difference contribution.

A.3 Kept-Error Control via Clopper–Pearson

Setup. Let $\mathcal{C} = \{(x_i, y_i)\}_{i=1}^n$ be an independent i.i.d. sample from the target distribution. Conditional on training and the separate score-fitting set D_{fit}, fix the calibrated class probabilities, cost-aware prediction $\hat{y}(x) = \arg\min_j e_j(x)$, score $s : \mathcal{X} \to \mathbb{R}$, and finite grid $\mathcal{T}$. Use deterministic tie-breaking. For threshold τ define

$$a_\tau(x) = \mathbf{1}\{\min_j e_j(x) \le c_{\text{abs}},\ s(x) \ge \tau\},$$
$$\mathcal{K}(\tau) = \{i : a_\tau(x_i) = 1\}, \qquad n_{\text{keep}}(\tau) = |\mathcal{K}(\tau)|.$$

With $M_i = \mathbf{1}\{\hat{y}(x_i) \ne y_i\}$, define

$$K(\tau) = \sum_{i \in \mathcal{K}(\tau)} M_i, \qquad \widehat{R}_{\text{keep}}(\tau) = \frac{K(\tau)}{n_{\text{keep}}(\tau)},$$

and define the population kept error by

$$R_{\text{keep}}(\tau) = \frac{\mathbb{E}[a_\tau(X)\mathbf{1}\{\hat{y}(X) \ne Y\}]}{\mathbb{E}[a_\tau(X)]}.$$

Both ratios use $0/0 = 0$. At positive coverage the population ratio equals $\Pr(\hat{y}(X) \ne Y \mid a_\tau(X) = 1)$.

Binomial model at a fixed threshold. For a fixed τ with positive population coverage, independence and identical sampling give

$$K(\tau) \mid n_{\text{keep}}(\tau) \sim \text{Binomial}\big(n_{\text{keep}}(\tau), \theta(\tau)\big), \qquad \theta(\tau) = R_{\text{keep}}(\tau).$$

Thus the kept errors are Bernoulli trials with parameter $\theta(\tau)$ conditional on which observations are kept.

Clopper–Pearson one-sided upper bound. For $q \in (0, 1)$ and integers $0 \le k \le n$, define

$$\text{CP}^+(k, n, q) = \begin{cases} \text{Beta}^{-1}(1 - q; k + 1, n - k), & 0 \le k < n, \\ 1, & k = n, \end{cases}$$

including $\text{CP}^+(0, 0, q) = 1$. For $k < n$, this inverts the lower binomial tail:

$$\Pr_{Z \sim \text{Binomial}(n, \text{CP}^+(k,n,q))}(Z \le k) = q.$$

Consequently, for any fixed τ,

$$\Pr\big(\theta(\tau) \le \text{CP}^+(K(\tau), n_{\text{keep}}(\tau), q)\big) \ge 1 - q. \tag{12}$$

For zero population coverage this holds under the stated zero-risk convention.

Threshold selection on a finite grid. Fix $\mathcal{T} = \{\tau_1 < \cdots < \tau_M\}$ before observing $\mathcal{C}$; for example, use score quantiles fitted on D_{fit}. Set $\delta' = \delta/M$ and

$$\tau^\star = \min\{\tau \in \mathcal{T} : \text{CP}^+(K(\tau), n_{\text{keep}}(\tau), \delta') \le \alpha\}.$$

If no grid threshold is feasible, set $\tau^\star = +\infty$ and abstain on all inputs.

Proposition 2 (Finite-sample kept-error control with grid search)**.** *With this construction,*

$$\Pr\left(R_{\text{keep}}(\tau^\star) \le \alpha\right) \ge 1 - \delta.$$

Proof. Conditional on the frozen model, score, and grid, apply (12) with $q = \delta'$ to each threshold. By the union bound,

$$\Pr\big(\forall \tau \in \mathcal{T} : \theta(\tau) \le \text{CP}^+(K(\tau), n_{\text{keep}}(\tau), \delta')\big) \ge 1 - M\delta' = 1 - \delta.$$

On this event a feasible selected threshold satisfies $\theta(\tau^\star) \le \text{CP}^+(K(\tau^\star), n_{\text{keep}}(\tau^\star), \delta') \le \alpha$. If no threshold is feasible, the abstain-all policy has risk zero under the stated convention. Averaging over the independent score-fitting data preserves the probability bound. □

A.4 EJOP via Finite Differences

Objects and estimator. Let $p^\star(x) \in [0, 1]^K$ denote the Bayes class probabilities, with coordinates $p_c^\star(x)$. Define the *EJOP* matrix

$$H_0 \ := \ \mathbb{E}\left[\sum_{c=1}^K \nabla p_c^\star(X)\, \nabla p_c^\star(X)^\top\right] \in \mathbb{R}^{d \times d}.$$

We estimate $\nabla p_c^\star$ by centered, coordinatewise finite differences applied to a surrogate $\tilde{p}_c$. For positive steps ε_j and unit coordinate vectors e_j, let $q_j^\pm = x \pm \varepsilon_j e_j$, $z_j^\pm = \Pi_Q(q_j^\pm)$, and

$$g_j(x;c) = \frac{\tilde{p}_c(z_j^+) - \tilde{p}_c(z_j^-)}{2\varepsilon_j}, \qquad g(x;c) = (g_1, \ldots, g_d)^\top.$$

Here Π_Q projects onto the empirical inter-quantile feature box; identity projection recovers unclipped probes. For a uniformly selected subsample $\{x_i\}_{i=1}^m$, the plug-in estimator is

$$\widehat{H}_0 \;\coloneqq\; \frac{1}{m}\sum_{i=1}^{m}\sum_{c=1}^{K} g(x_i;c)\, g(x_i;c)^\top.$$

Assumptions. We assume:

- (A1) **Smoothness.** Each $p_c^\star$ is continuously differentiable with L-Lipschitz gradient on a convex neighborhood containing the support of P_X, probe points, and clipped probe points: $\|\nabla p_c^\star(x) - \nabla p_c^\star(y)\|_2 \le L\|x-y\|_2$.
- (A2) **Surrogate accuracy.** $\|\tilde{p}_c - p_c^\star\|_\infty \le \eta$ on that neighborhood for every c.
- (A3) **Bounded gradients.** $\|\nabla p_c^\star(x)\|_2 \le B_{\max}$ throughout that neighborhood for every c.
- (A4) **Sampling.** The training inputs are i.i.d. from P_X, and the m distinct subsample indices are selected uniformly independently of their values.

Pointwise finite-difference error (discretization + surrogate). Write $\delta_c = \tilde{p}_c - p_c^\star$ and first decompose the unclipped finite difference $\bar{g}_j(x;c)$:

$$\bar{g}_j(x;c) \;=\; \underbrace{\frac{p_c^\star(x+\varepsilon_j e_j) - p_c^\star(x-\varepsilon_j e_j)}{2\varepsilon_j}}_{\text{centered difference on } p_c^\star} \;+\; \underbrace{\frac{\delta_c(x+\varepsilon_j e_j) - \delta_c(x-\varepsilon_j e_j)}{2\varepsilon_j}}_{\text{surrogate term}}.$$

Surrogate term. Since $|\delta_c(\cdot)| \le \eta$ pointwise,

$$\left|\frac{\delta_c(x+\varepsilon_j e_j) - \delta_c(x-\varepsilon_j e_j)}{2\varepsilon_j}\right| \;\le\; \frac{|\delta_c(x+\varepsilon_j e_j)| + |\delta_c(x-\varepsilon_j e_j)|}{2\varepsilon_j} \;\le\; \frac{\eta}{\varepsilon_j}.$$

Discretization term. By Taylor's theorem with remainder along $t \mapsto p_c^\star(x + te_j)$ and (A1),

$$\frac{p_c^\star(x+\varepsilon_j e_j) - p_c^\star(x-\varepsilon_j e_j)}{2\varepsilon_j} \;=\; \partial_{x_j} p_c^\star(x) \;+\; r_j(x;c),$$

with

$$|r_j(x;c)| \le \frac{1}{2\varepsilon_j}\int_{-\varepsilon_j}^{\varepsilon_j} |\partial_j p_c^\star(x+te_j) - \partial_j p_c^\star(x)|\, dt \le \frac{L}{2}\varepsilon_j.$$

(If $p_c^\star$ is C^3 with $\sup|\partial_{x_j}^3 p_c^\star| \le T_j$, one may strengthen to $|r_j| \le \frac{T_j}{6}\varepsilon_j^2$; we keep the $O(\varepsilon_j)$ bound implied by L-smoothness.)

For clipped probes, set $r_j^\pm(x) = \|z_j^\pm - q_j^\pm\|_2$ and define

$$b_Q(x) = \frac{B_{\max}}{2}\left[\sum_{j=1}^{d}\left(\frac{r_j^+(x) + r_j^-(x)}{\varepsilon_j}\right)^2\right]^{1/2}. \tag{13}$$

By (A3), changing the true-probability probes from $q_j^\pm$ to $z_j^\pm$ changes their finite difference by at most $B_{\max}(r_j^+ + r_j^-)/(2\varepsilon_j)$. The surrogate contribution at the clipped probes remains at most η/ε_j. Combining these terms and stacking over coordinates gives

$$\begin{aligned}\|g(x;c) - \nabla p_c^\star(x)\|_2 &\le C_1 \max_j \varepsilon_j + C_2\frac{\eta}{\min_j \varepsilon_j} + b_Q(x) =: \delta_g(x),\\ C_1 = \tfrac{L}{2}\sqrt{d}, &\qquad C_2 = \sqrt{d}.\end{aligned} \tag{14}$$

Outer-product perturbation from gradient error. For any $a, b \in \mathbb{R}^d$,

$$aa^\top - bb^\top \;=\; (a-b)b^\top + a(a-b)^\top,$$

so

$$\|\, aa^\top - bb^\top \,\|_{\mathrm{op}} \;\le\; \|a-b\|_2\,\|b\|_2 \;+\; \|a\|_2\,\|a-b\|_2 \;\le\; (\,\|a\|_2 + \|b\|_2\,)\,\|a-b\|_2. \tag{15}$$

Apply (15) with $a = g(x;c)$ and $b = \nabla p_c^\star(x)$ and use (A3):

$$\| g(x;c)g(x;c)^\top - \nabla p_c^\star(x)\nabla p_c^\star(x)^\top \|_{\mathrm{op}} \;\le\; \big(\|g(x;c)\|_2 + \|\nabla p_c^\star(x)\|_2 \big)\, \|g(x;c) - \nabla p_c^\star(x)\|_2.$$

Since $\|g(x;c)\|_2 \le \|\nabla p_c^\star(x)\|_2 + \|g - \nabla p_c^\star\|_2 \le B_{\max} + \Delta(x;c)$ with $\Delta(x;c) \coloneqq \|g - \nabla p_c^\star\|_2$, we obtain

$$\| gg^\top - \nabla p_c^\star \nabla p_c^{\star\top} \|_{\mathrm{op}} \;\le\; \big(2B_{\max} + \Delta(x;c)\big)\, \Delta(x;c) \;\le\; 2B_{\max}\Delta(x;c) + \Delta(x;c)^2. \tag{16}$$

Summing over c and averaging over the sample $\{x_i\}$ gives the *bias* term

$$\begin{aligned} \mathsf{Bias} \;&\coloneqq\; \left\| \frac{1}{m} \sum_{i=1}^{m} \sum_{c=1}^{K} \Big(g_i(c) g_i(c)^\top - \nabla p_c^\star(x_i) \nabla p_c^\star(x_i)^\top \Big) \right\|_{\mathrm{op}} \\ &\le\; \frac{1}{m} \sum_{i=1}^{m} \sum_{c=1}^{K} \big(2B_{\max}\Delta_i(c) + \Delta_i(c)^2 \big), \qquad \Delta_i(c) \;\coloneqq\; \|g(x_i;c) - \nabla p_c^\star(x_i)\|_2. \end{aligned} \tag{17}$$

By (14), deterministically $\Delta_i(c) \le \delta_g$ with

$$\delta_g \coloneqq \max_{1 \le i \le m} \delta_g(x_i) = C_1 \max_j \varepsilon_j + C_2 \frac{\eta}{\min_j \varepsilon_j} + \max_{1 \le i \le m} b_Q(x_i).$$

hence

$$\mathsf{Bias} \;\le\; K\big(2B_{\max}\delta_g + \delta_g^2\big). \tag{18}$$

Concentration for the sampling fluctuation (matrix Bernstein). Define the *oracle* summands

$$Y_i \;\coloneqq\; \sum_{c=1}^{K} \nabla p_c^\star(x_i)\, \nabla p_c^\star(x_i)^\top, \qquad \mathbb{E}[Y_i] = H_0,$$

and centered matrices $Z_i \coloneqq Y_i - \mathbb{E}[Y_i]$. By (A4), these oracle matrices are self-adjoint and independent, even when the surrogate and clipping box are estimated from the training sample. Each Y_i satisfies

$$\|Y_i\|_{\mathrm{op}} \;\le\; \sum_{c=1}^{K} \|\nabla p_c^\star(x_i)\|_2^2 \;\le\; KB_{\max}^2 \quad\Rightarrow\quad \|Z_i\|_{\mathrm{op}} \;\le\; \|Y_i\|_{\mathrm{op}} + \|H_0\|_{\mathrm{op}} \;\le\; 2KB_{\max}^2 \;=:\; R.$$

For the variance proxy,

$$\left\| \sum_{i=1}^{m} \mathbb{E}[Z_i^2] \right\|_{\mathrm{op}} \;\le\; \sum_{i=1}^{m} \mathbb{E}\|Z_i\|_{\mathrm{op}}^2 \;\le\; mR^2 \;=:\; \sigma^2.$$

Tropp's matrix Bernstein inequality yields, for any $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$\begin{aligned} \left\| \frac{1}{m} \sum_{i=1}^{m} Z_i \right\|_{\mathrm{op}} &\le \frac{\sqrt{2\sigma^2 \log(2d/\delta)}}{m} + \frac{2R \log(2d/\delta)}{3m} \\ &= 2KB_{\max}^2 \sqrt{\frac{2\log(2d/\delta)}{m}} + \frac{4KB_{\max}^2 \log(2d/\delta)}{3m}. \end{aligned} \tag{19}$$

We denote the RHS by $\mathsf{Conc}(m, \delta)$.

Putting it together (bias+concentration). Add and subtract the oracle mean:

$$\widehat{H}_0 - H_0 \;=\; \underbrace{\Big(\widehat{H}_0 - \tfrac{1}{m} \sum_i Y_i \Big)}_{\text{bias from } g \text{ vs. } \nabla p^\star} \;+\; \underbrace{\Big(\tfrac{1}{m} \sum_i Y_i - H_0 \Big)}_{\text{sampling fluctuation}}.$$

By (18) and (19), with probability at least $1 - \delta$,

$$\big\| \widehat{H}_0 - H_0 \big\|_{\mathrm{op}} \;\le\; K\big(2B_{\max}\delta_g + \delta_g^2\big) \;+\; 2KB_{\max}^2 \sqrt{\frac{2\log(2d/\delta)}{m}} \;+\; \frac{4KB_{\max}^2}{3}\, \frac{\log(2d/\delta)}{m}. \tag{20}$$

Remark 2 (Balancing ε_j against η)**.** *For a common step, the discretization–surrogate component $\delta_{\mathrm{FD}} = C_1\varepsilon + C_2\eta/\varepsilon$ is minimized at $\varepsilon^\star = \sqrt{2\eta/L}$ when $L, \eta > 0$. Our adaptive rule $\varepsilon_j = \max\{\alpha_\varepsilon \,\mathrm{MAD}(X_{:j})/0.6745, \varepsilon_{\mathrm{floor}}\}$ rescales positive steps to the feature scale while letting α_ε tune this tradeoff. The clipping contribution is given separately by* (13)*.*

Remark 3 (Ridge + trace normalization). *For the normalization in Sec. 2.4, let $\widehat{H} = d(\widehat{H}_0 + \gamma I)/T_{\widehat{H}_0}$ and $H = d(H_0 + \gamma I)/T_{H_0}$, where $T_A = \mathrm{tr}(A + \gamma I)$. For positive traces,*

$$\begin{aligned}\|\widehat{H} - H\|_{\mathrm{op}} &\le \frac{d\|\widehat{H}_0 - H_0\|_{\mathrm{op}}}{T_{H_0}} \\ &\quad + \frac{d\|\widehat{H}_0 + \gamma I\|_{\mathrm{op}}}{T_{\widehat{H}_0} T_{H_0}} \big|\,\mathrm{tr}(\widehat{H}_0 - H_0)\big| \\ &\le \left(\frac{d}{T_{\min}} + \frac{d^2\|\widehat{H}_0 + \gamma I\|_{\mathrm{op}}}{T_{\min}^2}\right) \|\widehat{H}_0 - H_0\|_{\mathrm{op}},\end{aligned}$$

where $T_{\min} = \min\{T_{\widehat{H}_0}, T_{H_0}\}$. Positive ridge gives $T_{\min} \ge \gamma d > 0$. Thus (20) *transfers to the normalized preconditioner with the displayed multiplicative factor.*

Corollary 2 (DIR-G proxy error). *Let $\ell(x) = \sum_{c=1}^{K} w_c p_c^\star(x)$ for fixed nonnegative cost weights w_c, and define $\tilde{g}(x) = \sum_c w_c g(x; c)$. The choice $w_c = C_{cj}$ gives the expected cost of a fixed prediction action j. By the triangle inequality and* (14)*,*

$$\|\,\tilde{g}(x) - \nabla\ell(x)\,\|_2 \;\le\; \sum_c w_c\,\|g(x;c) - \nabla p_c^\star(x)\|_2 \;\le\; \|w\|_1\,\delta_g(x).$$

Algorithmic Specification (Pseudocode)

B.1 End-to-End SCARF Pipeline

Algorithm 1 SCARF: Training and Inference

Require: Training data D_{tr}, independent score-fitting data D_{fit} and certification data D_{cal}, label set $\mathcal{Y}$ with K classes, cost matrix C, abstain cost c_{abs}, target α, failure probability δ, FD scale α_ε, positive step floor $\varepsilon_{\mathrm{floor}}$, EJOP size m, ridge $\gamma > 0$, DIR-G strength $\rho \ge 0$, forest configuration RF_cfg

Ensure: Forest F, feature map Φ, calibrated probabilities $\hat{p}$, score s, threshold $\tau^\star$

For each feature j, set $\varepsilon_j \leftarrow \max\{\alpha_\varepsilon \mathrm{MAD}(X_{:j})/0.6745, \varepsilon_{\mathrm{floor}}\}$.
Train a light surrogate $\tilde{p}$ on D_{tr}.
$\widehat{H}_0 \leftarrow$ EstimateEJOP$(\tilde{p}, \{\varepsilon_j\}, D_{\mathrm{tr}}, m)$ ▷ Alg. 2
$\widehat{H} \leftarrow d(\widehat{H}_0 + \gamma I)/\mathrm{tr}(\widehat{H}_0 + \gamma I)$; define $\Phi(x) \leftarrow \widehat{H}x$.
if $\rho > 0$ **then**
    $D_{\mathrm{tr}}^+ \leftarrow$ DirGAugment$(\tilde{p}, \{\varepsilon_j\}, D_{\mathrm{tr}}, \rho, C)$ ▷ Alg. 3
else
    $D_{\mathrm{tr}}^+ \leftarrow D_{\mathrm{tr}}$
end if
$F \leftarrow$ TrainForest$(\{(\Phi(x), y) : (x, y) \in D_{\mathrm{tr}}^+\}, \mathsf{RF_cfg})$.
$(\hat{p}, s, \mathcal{T}) \leftarrow$ CalibrateAndScore$(F, \Phi, D_{\mathrm{fit}}, C, c_{\mathrm{abs}})$ ▷ Alg. 4
Define $e_j(x) = \sum_{i \in \mathcal{Y}} \hat{p}_i(x) C_{ij}$, $\hat{y}(x) = \arg\min_j e_j(x)$ with fixed tie-breaking, and $a_0(x) = \mathbf{1}\{\min_j e_j(x) \le c_{\mathrm{abs}}\}$.
$\tau^\star \leftarrow$ CPThreshold$(s, \hat{y}, a_0, \mathcal{T}, D_{\mathrm{cal}}, \alpha, \delta)$ ▷ Alg. 5
For test input x, predict $\hat{y}(x)$ if $a_0(x) = 1$ and $s(x) \ge \tau^\star$; otherwise abstain (cost c_{abs}).

B.2 EJOP via Centered Finite Differences

Algorithm 2 ESTIMATEEJOP$(\tilde{p}, \{\varepsilon_j\}, D_{\mathrm{tr}}, m)$

Draw $m \le |D_{\mathrm{tr}}|$ distinct indices uniformly independently of their values to form $S = \{x_i\}_{i=1}^{m}$; define the empirical clipping box Q.
Initialize $\widehat{H}_0 \leftarrow 0_{d\times d}$.
for each $x \in S$ **do**
  for $c = 1$ to K **do**
    for $j = 1$ to d **do**
      $g_j(x;c) \leftarrow \dfrac{\tilde{p}_c(\Pi_Q(x+\varepsilon_j e_j)) - \tilde{p}_c(\Pi_Q(x-\varepsilon_j e_j))}{2\varepsilon_j}$
    end for
    $g(x;c) \leftarrow (g_1, \ldots, g_d)^\top$
    $\widehat{H}_0 \leftarrow \widehat{H}_0 + g(x;c)\, g(x;c)^\top$
  end for
end for
return $\widehat{H}_0/m$

B.3 Gradient-Aligned Data Augmentation (DIR-G)

Algorithm 3 DIRGAUGMENT$(\tilde{p}, \{\varepsilon_j\}, D_{\mathrm{tr}}, \rho, C)$

Define class weights $w_c \ge 0$ from the cost matrix C (e.g., $w_c = \sum_j C_{cj}$ or task-specific).
Initialize D^+ as an empty sequence.
for each $(x, y) \in D_{\mathrm{tr}}$ **do**
  for $c = 1$ to K **do**
    $g(x;c) \leftarrow 0_d$
    for each continuous feature j **do**
      $g_j(x;c) \leftarrow \dfrac{\tilde{p}_c(\Pi_Q(x+\varepsilon_j e_j)) - \tilde{p}_c(\Pi_Q(x-\varepsilon_j e_j))}{2\varepsilon_j}$
    end for
  end for
  $g(x) \leftarrow \sum_{c=1}^{K} w_c\, g(x;c)$ ▷ cost-aware surrogate gradient
  $u(x) \leftarrow g(x)/\|g(x)\|_2$ if $\|g(x)\|_2 > 0$; otherwise $u(x) \leftarrow 0$
  $x^{(+)} \leftarrow x + \rho\, u(x), \quad x^{(-)} \leftarrow x - \rho\, u(x)$
  Append $(x, y), (x^{(+)}, y), (x^{(-)}, y)$ to D^+, retaining repeated examples.
end for
return D^+

B.4 Calibration and Confidence Score

Algorithm 4 CALIBRATEANDSCORE$(F, \Phi, D_{\mathrm{fit}}, C, c_{\mathrm{abs}})$

For binary labels $\{0, 1\}$, fit isotonic $\varphi : [0, 1] \to [0, 1]$ to $\{(F_1(\Phi(x)), y) : (x, y) \in D_{\mathrm{fit}}\}$.
Define $\hat{p}_1(x) = \varphi(F_1(\Phi(x)))$, $\hat{p}_0(x) = 1 - \hat{p}_1(x)$; for $K > 2$, fit a probability calibrator with outputs in the probability simplex on D_{fit}.
Compute $e_j(x) = \sum_{i\in\mathcal{Y}} \hat{p}_i(x) C_{ij}$ and the margin, confidence, and severity components from Section 2.7.
Fit component means, scales, and severity reference statistics on D_{fit}; form the real-valued score $s(x)$ from Section 2.7.
Form a finite grid $\mathcal{T}$ from score quantiles on D_{fit}; remove duplicate thresholds.
Freeze and return $(\hat{p}, s, \mathcal{T})$.

B.5 Clopper–Pearson Threshold Selection

Algorithm 5 CPTHRESHOLD$(s, \hat{y}, a_0, \mathcal{T}, D_{\mathrm{cal}}, \alpha, \delta)$

Require: Frozen $s, \hat{y}, a_0, \mathcal{T}$, with $\mathcal{T} \neq \emptyset$, independent of the i.i.d. target-distribution sample D_{cal}.

$\delta' \leftarrow \delta/|\mathcal{T}|$; $\mathcal{F} \leftarrow \emptyset$.
for each $\tau \in \mathcal{T}$ **do**
  $\mathcal{K}(\tau) \leftarrow \{(x, y) \in D_{\mathrm{cal}} : a_0(x) = 1,\ s(x) \geq \tau\}$.
  $n_{\mathrm{keep}} \leftarrow |\mathcal{K}(\tau)|$; $K_{\mathrm{err}} \leftarrow \sum_{(x,y)\in\mathcal{K}(\tau)} \mathbf{1}\{\hat{y}(x) \neq y\}$.
  $\mathrm{UB} \leftarrow \mathrm{CP}^{+}(K_{\mathrm{err}}, n_{\mathrm{keep}}, \delta')$, with $\mathrm{CP}^{+}(n, n, \delta') = 1$, including $n = 0$.
  if $\mathrm{UB} \leq \alpha$ **then**
    $\mathcal{F} \leftarrow \mathcal{F} \cup \{\tau\}$.
  end if
end for
if $\mathcal{F} = \emptyset$ **then**
  return $\tau^{\star} \leftarrow +\infty$ ▷ abstain on all inputs
else
  return $\tau^{\star} \leftarrow \min \mathcal{F}$.
end if

Hyperparameters and Implementation Details

C.1 Global Configuration and Search Grids

Table 5: Global hyperparameters and search grids used across datasets. Paper defaults are in **bold**.

Component	Default	Grid / Options
Finite-difference step scale α_ε	**0.10**	$\{0.05,\ 0.10,\ 0.20\}$
EJOP subsample size m (fraction of train)	**1.0**	$\{0.3,\ 0.5,\ 1.0\}$
Ridge γ for preconditioner	**1e-3**	$\{1e{-}4,\ 1e{-}3,\ 1e{-}2\}$
Trace normalization	**on**	{on, off}
DIR-G strength ρ	**0.10**	$\{0,\ 0.05,\ 0.10,\ 0.20\}$
Forest (final) # trees	**500**	$\{300,\ 500,\ 800\}$
Forest (final) max depth	**None**	{None, 14, 20}
Forest (final) max features	$\sqrt{d}$	$\{\sqrt{d},\ d/3\}$
Forest (final) min samples leaf	**1**	$\{1,\ 3,\ 5\}$
Forest (surrogate) # trees	**200**	$\{100,\ 200\}$ (or multinomial logistic)
Calibration method	**Isotonic**	{Isotonic, Platt}
CP failure probability δ	**0.05**	$\{0.10,\ 0.05,\ 0.01\}$
Threshold grid size $\lvert\mathcal{T}\rvert$	**200**	$\{100,\ 200,\ 300\}$
Abstain cost c_{abs}	**task-specific**	Use task's cost model; sweeps reported in main text.

C.2 Training Configurations

Surrogate for EJOP/DIR-G. Random forest (learning_rate=0.05, max_depth=2, estimators=100). Class weights may mirror the application cost matrix.

Final forest. Unless otherwise stated, we use `n_estimators`=**500**, `max_features`=$\sqrt{d}$, `bootstrap`=**True**, `min_samples_leaf`=**1**, and leave `max_depth` unconstrained.

Calibration and scoring. Fit probability calibration and the standardized composite score of Section 2.7 on D_{fit}. Use the resulting calibrated probabilities for expected-cost minimization and freeze the score before CP certification on the independent D_{cal}.

C.3 CP Thresholding and Operating Points

Construct a grid $\mathcal{T}$ from **200** score quantiles on D_{fit}, removing duplicates. Scan this fixed real-valued grid on D_{cal} using Bonferroni-corrected one-sided Clopper–Pearson with $\delta' = \delta/|\mathcal{T}|$ and $\delta = 0.05$. For the reported operating points, $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. Count errors of the cost-minimizing classifier among points satisfying both the cost gate and score threshold. If no threshold has $\mathrm{CP}^+(K, n, \delta') \leq \alpha$, return the abstain-all policy.

C.4 Random Seeds, Versions, and Compute

Seeds and splits. We run **10** independent seeds $\{13, 37, 101, 202, 303, 404, 505, 606, 707, 808\}$. Each seed drives: (i) the train/cal/test split; (ii) subsample for EJOP; (iii) model initialization; (iv) any bootstrap sampling.

Table 6: Software versions and hardware.

Component	**Version / Spec**
OS	Ubuntu 22.04 LTS (x86_64)
Python	3.11.x
NumPy / SciPy	1.26.x / 1.11.x
scikit-learn	1.4.x
pandas	2.2.x
matplotlib (plots)	3.8.x
CPU	32 cores, 128 GB RAM
GPU (not required)	n/a

Code availability: https://drive.google.com/file/d/1tbjTpOqbiprukxzFdCqT50IqwsSdkjt4/view?usp=sharing